

Probability Theory Mastery Blueprint

A Structured Reference Guide to Set Axioms, Conditional Theorems, Combinatorics, and Distribution Functions

Probability theory provides the formal mathematical scaffold for modeling uncertainty, randomness, and risk. It serves as the baseline infrastructure for data analytics, machine learning algorithms, quantum mechanics, and statistical inference frameworks.

1. FOUNDATIONS, SAMPLE SPACES, & KOLMOGOROV AXIOMS

Let S represent the absolute **Sample Space** containing all possible outcomes of a random experiment, where an **Event (A)** is an explicit subset constraint ($A \subseteq S$).

Classical Definition: If all outcomes are equally likely, the probability of Event A is: $P(A) = n(A) / n(S)$
Where $n(A)$ is the number of favorable outcomes and $n(S)$ is the total sample cardinality.

Kolmogorov's Axioms of Probability

Every valid probability assignment must satisfy three fundamental axiomatic limits:

- Axiom 1 (Non-negativity):** For any event A, the boundary constraint holds: $P(A) \geq 0$.
- Axiom 2 (Certainty):** The probability of the entire sample space evaluates to absolute unity: $P(S) = 1$.
- Axiom 3 (Additivity):** For countably mutual, disjoint events ($A \cap B = \emptyset$), then: $P(A \cup B) = P(A) + P(B)$.

2. FUNDAMENTAL PROBABILITY THEOREMS & SET ALGEBRA

Set theory mappings dictate the calculations governing composite events across intersecting domains:

Theorem / Law Name	Mathematical Expression Matrix	Analytical Meaning
Complementary Rule	$P(A') = 1 - P(A)$	Probability of A not occurring.
General Addition Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Accounts for overlapping events.
Mutually Exclusive Rule	$P(A \cup B) = P(A) + P(B)$ (since $P(A \cap B) = 0$)	Events cannot happen together.
De Morgan's Mapping	$P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B)$	Probability that neither occurs.

3. CONDITIONAL PROBABILITY & EVENT INTERDEPENDENCY

Conditional probability evaluates the likelihood of an event occurring given that a prerequisite baseline condition has materialized.

Conditional Definition

The probability of Event A occurring *given* that Event B has already materialized is formulated as:

$$P(A | B) = P(A \cap B) / P(B)$$

Provided that base constraint satisfies $P(B) > 0$.

Independent Events

Two events are statistically independent if the materialization of one does not affect the likelihood of the other:

$$P(A \cap B) = P(A) \cdot P(B)$$

Yields structural equality: $P(A | B) = P(A)$.

4. LAW OF TOTAL PROBABILITY & BAYES' THEOREM

For a sample space partitioned completely into mutually exclusive, exhaustive chunks $B_1, B_2, \dots, B_n$:

The Law of Total Probability: Resolves a generic target event A across a scattered partition domain:

$$P(A) = \sum_{i=1}^n P(A | B_i) \cdot P(B_i)$$

Bayes' Theorem (Inference Framework): Calculates posterior belief updates by reversing dependencies:

$$P(B_k | A) = [P(A | B_k) \cdot P(B_k)] / [\sum_{i=1}^n P(A | B_i) \cdot P(B_i)]$$

5. COMBINATORICS: COUNTING AXIOMS & SAMPLE SELECTIONS

Combinatorial logic acts as the foundational calculation metric for establishing the cardinal sizes of finite sample structures:

Operation Model	Structural Formula	Core Structural Context Parameters
Permutations (Ordered)	${}^nP_r = n! / (n - r)!$	Arranging r items out of n distinct assets. Order matters.
Combinations (Unordered)	${}^nC_r = n! / [r!(n - r)!]$	Selecting a subset pool of r items where order is irrelevant.
Identical Items Layout	$N = n! / (n_1! \cdot n_2! \cdots n_k!)$	Permutations of items containing repeating identical subgroups.

6. RANDOM VARIABLES, EXPECTATIONS, & VARIANCE DYNAMICS

A random variable mapping function translates outcomes into quantitative real numerical coordinates.

Expected Value (Mean: μ) The long-term theoretical center of mass of a distribution function: • Discrete Variable: $E(X) = \sum x \cdot P(X = x)$ • Continuous Variable: $E(X) = \int_{-\infty}^{\infty} x \cdot f(x) \, dx$	Variance (Spread: σ^2) Measures the square dispersion boundary of coordinates about the mean: • Core Analytical Formula: $Var(X) = E(X^2) - [E(X)]^2$ • Standard Deviation (σ): $\sigma = \sqrt{Var(X)}$
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7. CLASSIC STANDARD PROBABILITY DISTRIBUTIONS MATRIX

Standardized templates for parametric uncertainty models across discrete and continuous bounds:

Distribution	Probability Mass/Density Function (PMF / PDF)	Mean ()	Variance ()
Binomial Discrete	$P(X = k) = {}^nC_k \cdot p^k \cdot (1 - p)^{n - k}$ Tracks k successes across n independent trials.	np	$np(1 - p)$
Poisson Discrete	$P(X = k) = (\lambda^k \cdot e^{-\lambda}) / k!$ Models independent event rates over fixed time intervals.	λ	λ
Exponential Continuous	$f(x) = \lambda \cdot e^{-\lambda x}$ (for $x \geq 0$) Models time durations preceding a sudden event breakthrough.	$1 / \lambda$	$1 / \lambda^2$
Normal Continuous	$f(x) = [1 / (\sigma\sqrt{2\pi})] \cdot \exp[-(x - \mu)^2 / (2\sigma^2)]$ Symmetric Gaussian bell curve dictated by the Central Limit Theorem.	μ	σ^2

8. THE CENTRAL LIMIT THEOREM (CLT)

The Central Limit Theorem forms the mathematical pillar of inferential analytics. It states that given a sufficiently large sample size ($n \geq 30$), the sampling distribution of the empirical sample mean ($\bar{X}$) will converge asymptotically toward a **Normal Gaussian Distribution**, completely independent of the shape or configuration of the parent distribution function:

$$X \sim N(\mu, \sigma^2 / n) \rightarrow Z = (X - \mu) / (\sigma / \sqrt{n})$$